

Reverse Engineering 1

SSEO TEAMS 2025								
TEAM #	SURNAME	SURNAME	SURNAME	SURNAME	SURNAME	SURNAME	SURNAME	marked in green who attends from remote
	1	2	3	4	5	6	7	
1	Bandello F.	Bagnoli A.	Balboni F.	Chico Alfonso S.	Pozzoni A.	Sagliani L.		LADEE
2	Borneo	Brusciani	Danile	Davi	De Vita	Di Santo		SDO
3	Atoche	Mahecha	Persico	Cavosi	Villari	Marie-Courtois		AEOLUS
4	Raymond	Richero	Bernasconi	Serlini	Palomba	Puzzi		SDO
5	Fossati	Mauroner	Gjoni	Jayakody Arachchilage	Fulgheri	Ghilaradini		AEOLUS
6	Bouvier J.	Temps R.	Beaujeu A.	Migeon M.	Ravet A.	Odeon V.		LADEE
7	Mauri	Luzzini	Di Giorgio	Perfetti	Scanga	Paiaardini		MRO
8	Amadei	Andreini	Dave	Difalco	Pirozzi	Preziosi		AEOLUS
9	Exner	Mages	Brun	Kilikauskas	Bouldoires	Jorgensen		LADEE
10	Macchi	Serlini P.	Roncetti	Formiconi	Sovani	Troggio	Oberti	AEOLUS
11	Pantoja Gavidia	Grijalva Medinilla	Paez Rioja	Comparato	D'Amico	Patruno		MRO
12	Andriollo	Bentini	Ciaci	Pirola	Vittore	Elfström		AEOLUS
13	Giannini	Frison	Ferrari	Emrem	Resta	Pirazzini		MRO
14	Maffioli	Loria	Grasso	Giaretta	Ottaviano	Ghelleri		AEOLUS
15	Cislaghi	Ahmed Mohamed	Ghoraba	Candita	Calderazzi	Giovenzana		LADEE
16	Addeo	Bove	Iaccarino	Marcolungo	Paripurnam	Pallotta		SDO
17	Cassano	Centrella	Burlando	Morelli	Brenna	Delpin	Harun Or	LADEE
18	Torrano	Balboni E.	Severo	Messina	Didonna	Burkey		MRO
19	Parma	Cavagna	Cammarota	Rizzi	Caucino	Bellezza		LADEE
20	Perniciaro	Pavan	Sarcina	Spitalieri	Parameswaran	Sousa		MRO
21	Chini	Mazzitti	Bianco	Basci	De Gennaro	Matteotti	Kazénas	MRO
22	Sorrieul	Deutsch	Kumar	Bredeau	Capano	Di Tolve		MRO
23	Erba	Fedele	Felice	Ferrari	Montemagno	Morrone		AEOLUS
24	Isernia	Bassani	Caridi	Mazza	Pracucci	Bisca		MRO
25	Monaco	Montepiani	Svizzero	Robotti	Melchiorre	Brackenier		LADEE
26	Balansard	De León	Frías Hernández	Garghetti	Herrera Medina	Rizzo		SDO
27	Genoni	Poletti	Sakova	Tizzi	Vicenzetto	Visintainer		MRO
28	Ruiu	Scoleri	Maggisano	Salvatori	Trevisan	Duranti		AEOLUS
29	Dolciotti	Chiappini	Bergamin	Ceola	Ruggiero	Matsui		SDO
30	Bacconier	Cocomazzi	Colombo A.	Coppola	Domenichelli	Pezzi		LADEE
31	Donati	Corbo	Chianese	Jonsson	Barin	Carminati		SDO
32	D'Onofrio	Lewis	Cicchetti	Musile Tanzi	Caironi	Capone		SDO
33	Negoita	Guercio	Ismailoglu	Sulis	Valvo	Digati		SDO
34	Capra	Bonati	Paniago	Caccia	Terzoli	Nasr		LADEE
35	Valsecchi	Sturiale	De Paoli	Falessi	Tajarollo	Esposito		AEOLUS
36	Barbera	Berzoini	Mariani	Muszynski	Boschini	Gonzalez		SDO

SSEO RE delivery timeline		
Homework #	Delivery deadline	Topic
1	March 16, 2025 - 23:30h	Obj, FUNC
2	March 30, 2025 - 23:30h	MA, PS
3	April 17, 2025 -23:30h	TTMTC
4	April 28, 2025 -23:30h	ADCS
5	May 11, 2025 -23:30h	TCS
6	May 18 2025 - 23:30h	EPS
7	May 31, 2025 - 23:30h	CONF\STR
8	June 8, 2025 - 23:30h	OBDH
9a	July 18, 2025 – 18:00h	FINAL REPORT
		1st delivery window
9b	September 7, 2025 – 18:00h	FINAL REPORT
		2nd delivery window

NEW

Missions – Homework 1

- Understand the mission high level goals
- Identify mission drivers, if any
- Perform the mission functional analysis
- Identify its main phases
- Link phases to ConOps and functionalities
- Fix the mission\system level requirements
- Understand the on board scientific instruments primary utilisation:
 - Correlate goal-to payloads functions
 - Correlate p\l to conops\phases
- Start correlating functionalities-phases trajectory design (MA understanding)

DELIVERY

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pp: 6 max

(front page, table of content, list of references, list of acronyms EXCLUDED from the pp counting)

Reverse Engineering 2

Missions – Homework 2_1

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#pp refers to the technical content only

- Understand and justify the mission trajectory design according to:
 - Functionalities and phases
 - ΔV budget per maneuver
- Compute and justify the results about the costs of, through reverse sizing:
 - Main orbital maneuvers $\rightarrow$ if the mission has complex interplanetary transfers\models (aerobraking, 3BP):
 - Aerobraking: quantify the ΔV gained with drag with energetic balance
 - Compute the ΔV for the E $\rightarrow$ M transfer with KM or perturbed, no 3BP modelling requested.
 - Overall station keeping (center of mass only)
- Reconstruct the ΔV budget breakdown table reporting, per manouver and expenditure, with proper margin, as needed:
 - A column of your computation
 - A column with the literature real costs

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#pp refers to the technical content only

- Understand the mission propulsion architecture
- Identify the solutions for **primary and secondary** propulsion and clarify the rationale for the adopted design
- Find the rationale for and justify the selected propulsion type and architecture according to:
 - Operations\phases (which thrusters to do which maneuvering)
 - ΔV budget breakdown
- Compute and justify, through reverse sizing, adopting proper margins:
 - Propellant selection and masses
 - Tanks sizing (propellant\pressurant), number, material adopted
 - Pressurant selection and masses
 - Feeding strategy selection and sizing
 - # thrusters
 - Positioning of thrusters in the configuration
 - Positioning of tanks and lines in the configuration
 - PS schematics

Reverse Engineering 3

- Understand the mission TMTC architecture
- Identify the solutions for TMTC subsystem and clarify the rationale for the adopted design
- Find the rationale for and justify the selected TMTC type and architecture according to:
 - Operations\phases\modes
 - Data volume to transfer space2ground, ground2space
 - Ground segment network selection
- Understand and justify, through reverse sizing:
 - Frequency selection, data rate, band
 - Signal manipulation (encoding, modulation, etc)
 - Antenna selection, type, characteristics and numbers correlated to which frequency and for which data transfer
 - Ground station selection: where, which size\frequency\data rate
 - Amplifier selection
 - Contact strategy: contact windows duration and data volume transferred per contact (average)
 - Link budget Uplink\Downlink
 - Positioning of the antennae the configuration on the space segment
 - Subsystem architecture

Reverse Engineering 4

Missions – Homework 4

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#pp refers to the technical content only

- Understand the mission AOCS architecture
- Identify the solutions for AOCS subsystem and clarify the rationale for the adopted design
- Find the rationale for and justify the selected AOCS type and architecture according to:
 - Control mode
 - Pointing budget
- Understand and justify, through reverse sizing:
 - Pointing budgets inputs for each subsystem involved: AKE, APE
 - Attitude sensor suite selection according to mode – redundancy included
 - Attitude actuator suite selection according to mode
 - Attitude sensor sizing according to pointing knowledge needs, per mode
 - Disturbances effects, slew maneuvers, per mode\phase
 - Attitude actuator sizing according to disturbances and requested slew maneuvers per mode (torque, angular momentum, thrust, dipole, etc) – redundancy included
 - Fuel mass sizing according to: attitude SK, maneuvers, desaturation, if any
 - Subsystem budgets: mass, power, data
 - Positioning of the sensors and actuators in the s/c configuration